MA411 Notes 3

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8 Concentration of Measure

Given a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, and a real-valued random variable X on Ω, in general there is no particular reason why X is 'likely' to be 'close to' $\mathbb{E}(X)$: indeed, the expectation might not even exist.

If it does exist, and is non-negative, Markov's inequality gives us a weak bound like this.

If both first and second moment exist, Chebyshev's inequality gives us a bit more.

However many random variables of interest are sums of simpler random variables. Such random variables tend to *concentrate*: they are 'very likely' to be 'close' to their expectation.

A detailed quantitative version of this is the following famous theorem.

Theorem 1 (Central Limit Theorem)**.** *Let $Y_1, \dots$ be independent, identically distributed random variables on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, such that $\mathbb{E}(Y_i) = \mu$ and $\mathrm{Var}(Y_i) = \sigma^2$ are real numbers. Let $X_n = \sum_{i=1}^n Y_i$. Then we have, for any $c \in \mathbb{R}$:*

$$\lim_{n\to\infty} \mathbb{P}\Big(\tfrac{X_n - \mu n}{\sigma\sqrt{n}} \in (-\infty, c)\Big) = \mathbb{P}\big(N(0,1) \in (-\infty, c)\big)\,.$$

We won't prove this. In this course, we are not so interested in the precise 'shape' of the distribution in the limit. In fact, we aren't even interested in such a limit existing. Also, we aren't told 'how fast' convergence occurs in the above (though there are versions of the CLT that come with such guarantees!) and this is something we *do* want to know.

What we would like to know is strong quantitative bounds, ideally that are 'easy to use', on the *tails* of the distribution: what are good upper bounds on $\mathbb{P}\big(X_n \geq \mathbb{E}(X_n) + t_n\big)$ and $\mathbb{P}\big(X_n \leq \mathbb{E}(X_n) - t_n\big)$? In both cases, we're thinking of t_n as being the same order of magnitude as $\mathbb{E}(|X_n|)$ in most applications.

Theorem 2 (Hoeffding's inequality)**.** *Let $Y_1, \dots, Y_n$ be independent random variables on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Suppose that almost surely $a_i \leq Y_i \leq b_i$ for all $1 \leq i \leq n$, where $a_i, b_i \in \mathbb{R}$. Let $X = \sum_{i=1}^n Y_i$. Then we have*

$$\mathbb{P}\big(X - \mathbb{E}(X) \geq t\big), \mathbb{P}\big(X - \mathbb{E}(X) \leq -t\big) \leq \exp\Big(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\Big)\,.$$

We didn't here explicitly ask for $\mathbb{E}(|Y_i|) < \infty$, but the condition that it's almost surely bounded in particular gives that this expectation is at most $\max(|a_i|, |b_i|)$.

You should notice that the probability bound here looks 'a bit like' the bound in a Gaussian random variable: one should think of $(b_i - a_i)^2$ as being an upper bound for $\mathrm{Var}(Y_i)$ (which is what would appear if we summed independent Gaussian random variables). But this bound is 'easy to use' in that we have a simple formula that we can just plug t in to.

We'll prove this theorem in two steps (and reuse one step later).

Lemma 3. *Let $Y_1, \ldots, Y_n$ be independent random variables on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Suppose that $\mathbb{E}(Y_i) = 0$, and almost surely $a_i \leq Y_i \leq b_i$, for all $1 \leq i \leq n$, where $a_i, b_i \in \mathbb{R}$. Let $X = \sum_{i=1}^n Y_i$. Then for all $s > 0$ we have*

$$\mathbb{E}\big(e^{sX}\big) \leq \exp\left(\frac{s^2}{8}\sum_{i=1}^n (b_i - a_i)^2\right).$$

Exercise 1. *Prove this. Hint: write $e^{sX} = e^{sY_1}e^{sY_2}\ldots e^{sY_n}$, use the reverse Jensen inequality, and optimise.*

Solution to Exercise 1:

Observe that $y \to e^{sy}$ is convex on $\mathbb{R}$, since $s > 0$. By the reverse Jensen inequality, we have

$$\mathbb{E}\big(e^{sY_i}\big) \leq \alpha e^{sa} + (1-\alpha)e^{sb}$$

where $\alpha a + (1-\alpha)b = 0$, so $\alpha = \frac{b}{b-a}$. Let $d = b - a$, and we get

$$\mathbb{E}\big(e^{sY_i}\big) \leq \tfrac{a+d}{d}e^{sa} - \tfrac{a}{d}e^{s(a+d)} = e^{g(s)},$$

where $g(s) = sa + \log\big(\frac{a+d}{d} - \frac{a}{d}e^{sd}\big)$. We compute derivatives:

$$g'(s) = a - \tfrac{a}{d}de^{sd}\cdot\frac{1}{\frac{a+d}{d} - \frac{a}{d}e^{sd}} \quad \text{and} \quad g''(s) = -a\frac{de^{sd}\big(\frac{a+d}{d} - \frac{a}{d}e^{sd}\big) + e^{sd}ae^{sd}}{\big(\frac{a+d}{d} - \frac{a}{d}e^{sd}\big)^2} = d^2\frac{(a+d)(-ae^{sd})}{\big((a+d) - ae^{sd}\big)^2}$$

We note $g(0) = g'(0) = 0$, and we use the inequality $4uv \leq (u+v)^2$ with $u = a + d$ and $v = -ae^{sd}$ to obtain $g''(s) \leq \frac{1}{4}d^2$ for all s. Integrating back, we have $g(s) \leq \frac{1}{8}d^2s^2$ for all s, and in particular

$$\mathbb{E}\big(e^{sY_1}\big) \leq e^{s^2(b_1-a_1)^2/8}.$$

The statement of the exercise now follows by multiplicativity of the expectation: since $Y_1, \ldots, Y_n$ are independent we have

$$\mathbb{E}\big(e^{sX}\big) = \mathbb{E}\big(e^{sY_1}\big)\mathbb{E}\big(e^{sY_2}\big)\ldots\mathbb{E}\big(e^{sY_n}\big) = \exp\left(\tfrac{s^2}{8}\sum_{i=1}^n (b_i - a_i)^2\right).$$

What goes with this is a general statement that an upper bound on the *moment generating function* $\mathbb{E}(e^sX)$ of X implies that X is unlikely to be too much greater than its expectation.

Lemma 4. *Let X be a random variable on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Suppose that for some $\lambda > 0$ we have for all $s > 0$ the bound*

$$\mathbb{E}(e^{sX}) \leq \exp\Big(s\mathbb{E}(X) + \lambda s^2\Big).$$

Then $\mathbb{P}\big(X \geq \mathbb{E}(X) + t\big) \leq \exp\big(-\frac{t^2}{4\lambda}\big)$.

Exercise 2. *Prove this. Hint: use Markov's inequality and optimise.*

Solution to Exercise 2:

For any $s > 0$, we can apply Markov's inequality to e^{sX}, which is a non-negative random variable. We note $x \to e^{sx}$ is increasing, to obtain

$$\mathbb{P}\big(X \geq \mathbb{E}(X) + t\big) = \mathbb{P}\Big(e^{sX} \geq e^{s\mathbb{E}(X)+st}\Big) \leq \mathbb{E}\big(e^{sX}\big)e^{-s\mathbb{E}(X)-st} \leq \exp\Big(\lambda s^2 - st\Big).$$

We would like to choose s to make the right side as small as possible: the right choice is $s = \frac{t}{2\lambda}$, so we get

$$\mathbb{P}\big(X \geq \mathbb{E}(X) + t\big) \leq \exp\Big(\tfrac{t^2}{4\lambda} - \tfrac{t^2}{2\lambda}\Big) = \exp\Big(-\tfrac{t^2}{4\lambda}\Big).$$

We can now prove Theorem 2.

Proof of Theorem 2. For any given $s > 0$, we apply Lemma 3 with the random variables $Y_i - \mathbb{E}(Y_i)$, which have expectation 0 by definition, and which are a.s. between $a_i - \mathbb{E}(Y_i)$ and $b_i - \mathbb{E}(Y_i)$, to obtain

$$\mathbb{E}\big(e^{s(X-\mathbb{E}(X))}\big) \leq \exp\Big(\tfrac{s^2}{8}\sum_{i=1}^{n}(b_i - a_i)^2\Big).$$

We conclude

$$\mathbb{E}\big(e^{sX}\big) \leq \exp\Big(s\mathbb{E}(X) + \tfrac{s^2}{8}\sum_{i=1}^{n}(b_i - a_i)^2\Big),$$

so we can apply Lemma 4 with $\lambda = \frac{1}{8}\sum_{i=1}^{n}(b_i - a_i)^2$, and we obtain

$$\mathbb{P}\big(X \geq \mathbb{E}(X) + t)\big) \leq \exp\Big(-\frac{2t^2}{\sum_{i=1}^{n}(b_i - a_i)^2}\Big).$$

Applying this to $-X$, we obtain the same bound on

$$\mathbb{P}\big(-X \geq -\mathbb{E}(X) + t\big) = \mathbb{P}\big(X \leq \mathbb{E}(X) - t\big). \qquad \square$$

This result (and relatives) is used a lot in discrete probability, in particular in combination with the union bound. A 'canonical' application is to come up with a long list of random variables, each of which can be written as a sum of independent bounded random variables. We apply Hoeffding's inequality to each in turn to show that it is very likely to be 'close to expectation', and then the union bound to argue that it is still quite likely that the entire list of random variables are all close to their expectations: and this is supposed to be 'what we wanted'.

Here is one example: we need a different concentration inequality, but the idea is otherwise the same.

Theorem 5 (Johnson-Lindenstrauss projection)**.** *Let* $\mathbf{v}$ *be any vector in* $\mathbb{R}^n$*, and let* $0 < \varepsilon \leq \frac{1}{2}$*. Let* A *be a* d *by* n *matrix, the entries of which are independent samples from the distribution* $N\big(0, \frac{1}{d}\big)$*.*

With probability at least $1 - 2\exp(-\varepsilon^2 d/8)$*, we have* $(1-\varepsilon)\|\mathbf{v}\|_2 \leq \|A\mathbf{v}\|_2 \leq (1+\varepsilon)\|\mathbf{v}\|_2$*.*

Proof. The theorem is trivial if $\mathbf{v} = \mathbf{0}$ (any matrix works!), so suppose this is not the case. We can scale $\mathbf{v}$ to assume $\|\mathbf{v}\|_2 = 1$: our desired conclusion isn't affected by scaling.

We want to do something like the previous proof: compute the moment generating function, then use Markov's inequality.

Something which hugely simplifies this is a nice property of Gaussian random variables. Let Y_i be the random variable which is the ith entry of $A\mathbf{v}$. Then we have $Y_i = \sum_{j=1}^n v_j X_j$ where the X_j are independent $N(0, d^{-1})$-random variables, and (by assumption) $\sum_{j=1}^n v_j^2 = 1$. It is a standard exercise to prove that Y_i is distributed as $N(0, d^{-1})$.

Exercise 3. *Prove that if* $Y = \sum_{j=1}^n \alpha_j Z_j$*, where the* Z_j *are independent* $N(0,1)$*-random variables, then* Y *is distributed as* $N\big(0, \sum_{j=1}^n \alpha_j^2\big)$*.*

Solution to Exercise 3:

By induction, it's enough to prove this statement for summing two independent Gaussian random variables. It's convenient to start with a general fact: if X, X' are two random variables which have probability density functions f, g respectively, then the probability density function of $X + X'$, evaluated at x, is $\int_{t=-\infty}^{\infty} f(x-t)g(t)\,\mathrm{d}t$. Cheating a little bit, this is proved as follows.

Given x, $n \in \mathbb{N}$ and $m \in \mathbb{Z}$, let

$$E_{n,m} = \big\{(y, y') \,:\, y < x - 2^{-n}m, y' \in [2^{-n}(m-1), 2^{-n}m)\big\}\,.$$

By independence, $\mathbb{P}((X, X') \in E_{n,m}) = \mathbb{P}(X < 2^{-n}m)\mathbb{P}(X' \in [2^{-n}(m-1), 2^{-n}m))$, and we can write the latter as a product of integrals. Let $E_n = \bigcup_{n,m} E_{n,m}$ which is a disjoint union, and we obtain

$$\mathbb{P}\big((X, X') \in E_n\big) = \int_{\mathbb{R}^2} \mathbf{1}_{E_n} f(y)g(y')\,\mathrm{d}(y, y')\,.$$

Observe that if, for a given $\omega \in \Omega$, we have $X + X' < x$ then there exists n such that $(X, X') \in E_n$, while if $X + X' \geq n$ then $(X, X') \notin E_n$. Furthermore, $E_1 \subseteq E_2 \subseteq \ldots$, so

$$\mathbb{P}(X+X' < x) = \lim_{n\to\infty} \mathbb{P}((X, X') \in E_n) = \lim_{n\to\infty} \int_{\mathbb{R}^2} \mathbf{1}_{E_n} f(y)g(y')\,\mathrm{d}(y, y') = \int_{\mathbb{R}^2} \mathbf{1}_{y+y'<x} f(y)g(y')\,\mathrm{d}(y, y')$$

by continuity of measure and the Monotone Convergence Theorem.

Now consider the change of variable $y = t' - t$, $y' = t$. We get

$$\int_{\mathbb{R}^2} \mathbf{1}_{y+y'<x} f(y)g(y')\,\mathrm{d}(y, y') = \int_{\mathbb{R}^2} \mathbf{1}_{t'<x} f(t'-t)g(t)\,\mathrm{d}(t, t') = \int_{t'=-\infty}^{x} \int_{t=-\infty}^{\infty} f(t'-t)g(t)\,\mathrm{d}t\,\mathrm{d}t'$$

which verifies that the given formula is a probability density function.

The 'cheating' here is that we assumed we can swap between integrals over $\mathbb{R}^2$ and iterated integrals: this is really by Fubini's Theorem. The change of variable, on the other

hand, is not cheating: it follows from Notes 1 that we obtain the same measure on $\mathbb{R}^2$ from both axis-parallel boxes and the skew boxes arising from the change of variable, so the two integrals coincide.

Suppose now $\alpha, \beta > 0$.

Observe

$$\mathbb{P}(\alpha Z < x) = \int_{t=-\infty}^{\alpha^{-1}x} \tfrac{1}{\sqrt{2\pi}} \exp\big(-\tfrac{t^2}{2}\big)\,\mathrm{d}t = \int_{t'=-\infty}^{x} \tfrac{1}{\alpha\sqrt{2\pi}} \exp\big(-\tfrac{(t')^2}{2\alpha^2}\big)\,\mathrm{d}t'$$

using the change of variable $t = t'/\alpha$. This shows αZ is distributed as $N(0, \alpha^2)$.

Now we just have to compute an integral. Suppose $Y = \alpha Z + \beta Z'$, where Z, Z' are distributed as $N(0,1)$. Then

$$\int_{t=-\infty}^{\infty} \tfrac{1}{2\pi\alpha\beta} e^{-(x-t)^2/(2\alpha^2)} e^{-t^2/(2\beta^2)}\,\mathrm{d}t = \int_{t=-\infty}^{\infty} \tfrac{1}{2\pi\alpha\beta} \exp\big(-\tfrac{1}{2}\tfrac{\beta^2(x-t)^2+\alpha^2t^2}{\alpha^2\beta^2}\big)\,\mathrm{d}t$$

Let $\gamma = \sqrt{\alpha^2+\beta^2}$. We complete the square in the exponent:

$$\beta^2(x-t)^2 + \alpha^2t^2 = (\alpha^2+\beta^2)t^2 - 2\beta^2tx + \beta^2x^2 = \gamma^2(t-\beta^2\gamma^{-2}x)^2 + \gamma^{-2}\alpha^2\beta^2x^2\,.$$

Plugging this in, we have

$$\begin{aligned}
\int_{t=-\infty}^{\infty} \tfrac{1}{2\pi\alpha\beta} \exp\big(-\tfrac{1}{2}\tfrac{\beta^2(x-t)^2+\alpha^2t^2}{\alpha^2\beta^2}\big)\,\mathrm{d}t &= \int_{t=-\infty}^{\infty} \tfrac{1}{2\pi\alpha\beta} \exp\big(-\tfrac{1}{2}\tfrac{\gamma^2(t-\beta^2\gamma^{-2}x)^2+\gamma^{-2}\alpha^2\beta^2x^2}{\alpha^2\beta^2}\big)\,\mathrm{d}t \\
&= \int_{t=-\infty}^{\infty} \tfrac{1}{2\pi\alpha\beta} \exp\big(-\tfrac{1}{2}\tfrac{\gamma^2(t-\beta^2\gamma^{-2}x)^2}{\alpha^2\beta^2}\big) e^{\frac{-x^2}{2\gamma^2}}\,\mathrm{d}t \\
&= \tfrac{1}{\sqrt{2\pi}} e^{\frac{-x^2}{2\gamma^2}} \int_{t=-\infty}^{\infty} \tfrac{1}{\sqrt{2\pi}\alpha\beta} \exp\big(-\tfrac{1}{2}\tfrac{\gamma^2(t-\beta^2\gamma^{-2}x)^2}{\alpha^2\beta^2}\big)\,\mathrm{d}t \\
&= \tfrac{1}{\sqrt{2\pi}\gamma} e^{\frac{-x^2}{2\gamma^2}} \int_{t=-\infty}^{\infty} \tfrac{\gamma}{\sqrt{2\pi}\alpha\beta} \exp\big(-\tfrac{1}{2}\tfrac{\gamma^2t^2}{\alpha^2\beta^2}\big)\,\mathrm{d}t \\
&= \tfrac{1}{\sqrt{2\pi}\gamma} e^{\frac{-x^2}{2\gamma^2}}
\end{aligned}$$

as required.

This means we don't need to worry any more about what exactly $\mathbf{v}$ is, or the matrix multiplication: we are just looking at a vector of d independent $N(0, d^{-1})$-random variables.

Exercise 4. *Prove that if $2t < d$ then $\mathbb{E}\big(e^{tY_1^2}\big) = \big(1 - \frac{2t}{d}\big)^{-1/2}$. Hint: write the integral out explicitly and use a change of variable to convert it into the Gaussian integral.*

Solution to Exercise 4:

Using the LOTUS, we calculate (on the assumption $2t/d < 1$, i.e. $t < d/2$)

$$\mathbb{E}\big[e^{tY_1^2}\big] = \int_{y=-\infty}^{\infty} \frac{\sqrt{d}}{\sqrt{2\pi}} e^{ty^2} e^{-y^2d/2}\mathrm{d}y = \int_{y=-\infty}^{\infty} \frac{\sqrt{d}}{\sqrt{2\pi}} e^{-(1-2t/d)y^2d/2}\mathrm{d}y\,,$$

and we use the change of variable $z^2 = (1-2t/d)y^2$ (this is where we use $2t/d < 1$), for which we have $\frac{\mathrm{d}y}{\mathrm{d}z} = (1-2t/d)^{-1/2}$. We get

$$\mathbb{E}\big[e^{tY_1^2}\big] = \int_{z=-\infty}^{\infty} \frac{\sqrt{d}}{\sqrt{2\pi}} e^{-z^2 d/2} \cdot (1-2t/d)^{-1/2} \mathrm{d}z = (1-2t/d)^{-1/2}\,,$$

where the final equality uses that the Gaussian integral evaluates to 1.

We now can compute the moment generating function $\mathbb{E}\big(e^{t\|A\mathbf{v}\|_2^2}\big) = \big(1-\frac{2t}{d}\big)^{-d/2}$ by multiplicativity of expectation for independent random variables. Since we are assuming $\|\mathbf{v}\|_2^2 = 1$, what we are trying to prove is that the probability of

$$(1-\varepsilon)^2 \le \|A\mathbf{v}\|_2^2 = Y_1^2 + \cdots + Y_d^2 \le (1+\varepsilon)^2$$

failing is at most $2e^{-\varepsilon^2 d/8}$. We'll show the two inequalities, separately, are unlikely to fail: we start with the upper bound.

We are now in a position to copy the proof of the Hoeffding inequality. We begin with, for any $0 \le t < d/2$, the observation

$$\mathbb{P}\big[Y_1^2 + \cdots + Y_d^2 > (1+\varepsilon)^2\big] = \mathbb{P}\big[e^{t(Y_1^2+\cdots+Y_d^2)} > e^{t(1+\varepsilon)^2}\big] \le \frac{\mathbb{E}\big[e^{t(Y_1^2+\cdots+Y_d^2)}\big]}{e^{t(1+\varepsilon)^2}} = \frac{(1-2t/d)^{-d/2}}{e^{t(1+\varepsilon)^2}}\,,$$

where the inequality is Markov. We need to optimise this over choices of t; a choice that works is $t = \frac{d}{2} \cdot \frac{2\varepsilon+\varepsilon^2}{(1+\varepsilon)^2}$, which is between 0 and $d/2$ as required. Substituting this in, we get

$$\mathbb{P}\big[Y_1^2 + \cdots + Y_d^2 > (1+\varepsilon)^2\big] \le \frac{(1-2t/d)^{-d/2}}{e^{t(1+\varepsilon)^2}} = \frac{(1+\varepsilon)^d}{e^{d(2\varepsilon+\varepsilon^2)/2}} < e^{-\varepsilon^2 d/8}\,,$$

which is as desired. The final inequality requires a bit of justification: probably the easiest way is to use the estimate $1+\varepsilon < e^{\varepsilon}$ (this is just the first two terms of the Taylor series), valid since $\varepsilon > 0$; we get

$$\frac{(1+\varepsilon)^d}{e^{d(2\varepsilon+\varepsilon^2)/2}} \le e^{\varepsilon d} e^{-\varepsilon d - \varepsilon^2 d/2} = e^{-\varepsilon^2 d/2} < e^{-\varepsilon^2 d/8}\,.$$

This proves the failure probability for the upper bound is at most $e^{-\varepsilon^2 d/8}$.

Exercise 5. *Prove that*

$$\mathbb{P}\Big((1-\varepsilon)^2 > \sum_{i=1}^{d} Y_i^2\Big) < e^{-\varepsilon^2 d/8}\,.$$

Hint: the moment generating function computation we did is also valid if $t < 0$.

Solution to Exercise 5:

Since $x \to e^{-tx}$ is a monotone decreasing function, we have

$$\mathbb{P}\Big((1-\varepsilon)^2 > \sum_{i=1}^{d} Y_i^2\Big) = \mathbb{P}\Big(\exp\big(-t\sum_{i=1}^{d} Y_i^2\big) > e^{-t(1-\varepsilon^2)}\Big) \le \mathbb{E}\Big(\exp\big(-t\sum_{i=1}^{d} Y_i^2\big)\Big) e^{t(1-\varepsilon)^2}$$

Substituting in our calculated moment generating function, we have

$$\mathbb{P}\Big((1-\varepsilon)^2 > \sum_{i=1}^{d} Y_i^2\Big) \leq \big(1+\tfrac{2t}{d}\big)^{-d/2} e^{t(1-\varepsilon)^2}$$

and choosing $t = \frac{d}{2}\frac{2\varepsilon-\varepsilon^2}{(1-\varepsilon)^2}$ we get

$$\mathbb{P}\Big((1-\varepsilon)^2 > \sum_{i=1}^{d} Y_i^2\Big) \leq \big(1+\tfrac{2\varepsilon-\varepsilon^2}{(1-\varepsilon)^2}\big)^{-d/2} e^{d(2\varepsilon-\varepsilon^2)/2} \leq e^{-\varepsilon^2 d/8}\,.$$

By the union bound, with probability at least $2e^{-\varepsilon^2 d/8}$, both the upper and lower bounds hold, as required. □

We can use this to prove the following:

Theorem 6 (Johnson-Lindenstrauss Theorem)**.** *Given $0 < \varepsilon \leq \frac{1}{2}$, n and d, let S be any set of at most $e^{\varepsilon^2 d/32}$ points in $\mathbb{R}^n$. There exists a d by n matrix A such that the following holds. For any two points $\mathbf{v}, \mathbf{v}' \in S$ we have*

$$(1-\varepsilon)\|\mathbf{v}-\mathbf{v}'\|_2 \leq \|A\mathbf{v} - A\mathbf{v}'\|_2 \leq (1+\varepsilon)\|\mathbf{v}-\mathbf{v}'\|_2\,.$$

Proof. There are at most $|S|^2$ pairs of points in S, so the set $S' = \{\mathbf{v}-\mathbf{v}' : \mathbf{v}, \mathbf{v}' \in S\}$ has size at most $|S|^2$. For each $\mathbf{v} \in S'$, let $E_{\mathbf{v}}$ be the event that $(1-\varepsilon)\|\mathbf{v}\|_2 \leq \|A\mathbf{v}\|_2 \leq (1+\varepsilon)\|\mathbf{v}\|_2$ fails.

By Theorem 5, for each $\mathbf{v} \in S'$ we have $\mathbb{P}(E_{\mathbf{v}}) \leq 2e^{-\varepsilon^2 d/8}$. By the union bound, we have

$$\mathbb{P}\Big(\bigcup_{\mathbf{v}\in S'} E_{\mathbf{v}}\Big) \leq |S|^2 \cdot 2e^{-\varepsilon^2 d/8} \leq 2e^{-\varepsilon^2 d/16} < 1\,,$$

and so with positive probability the random matrix A has the required property. In particular, there must exist an A which has the required property. □

This is useful in data science: if we have a lot of high-dimensional data (we always do!), we will want to process it in some way to find 'clusters' or otherwise try to understand the structure. Think of the following data set: every Amazon customer has a complete record of what items on Amazon, and how often, they bought. So every customer is a (huge!) vector of non-negative integers, of dimension equal to the number of products sold on Amazon: this is the high-dimensional data set.

Algorithms exist to do this. But these algorithms usually take longer, sometimes much longer, to run if the dimension is high. What the Johnson-Lindenstrauss Theorem says is that we can randomly project to $\mathbb{R}^d$ without changing pointwise distances much: we can run our algorithms on the projected data much more quickly, and (because in the end the pairwise distances are the structure!) the results will still be good.

While the theorem just says the desired matrix exists, the proof actually shows a random matrix is incredibly likely to be the matrix we want. In 'realistic' applications we would

allow for at least 10^9 data points, and even to apply the theorem, our failure probability is smaller than 10^{-18}. For safety we'd probably want to choose d perhaps 4 times as large as we 'need', and our failure probability drops to 10^{-72}. It's pretty safe to just hope it works: the chance of an asteroid impact taking out your computer is higher than of accidentally picking a 'bad' matrix. It's also 'safe' to generate the matrix ahead of time and then collect data (in which case you can even safely store only the 'compressed' projected data!), provided you won't ever collect more data than the theorem allows.

The only problem with this is that it is hard to generate Gaussian-distributed random numbers, and computations with high-precision real numbers are slow. We would really like a 'better' random matrix whose entries are easier to generate and work with.

Exercise 6. *In this exercise we try to show that the Johnson-Lindenstrauss Theorem also goes through for the following 'binary' random matrix. Let W be an n by d random matrix whose entries are independent and chosen from $\{-d^{-1/2}, d^{-1/2}\}$ with equal probability.*

(a) Verify that the entries of W have mean 0 and variance d^{-1}.

Fix $\mathbf{x} \in \mathbb{R}^n$ with $\|\mathbf{x}\|_2 = 1$.

(b) Calculate explicitly

$$\mathbb{E}\Big(\Big(\sum_{j=1}^{n} x_j W_{1j}\Big)^k\Big),$$

i.e. the expectation of $\big(W\mathbf{x}\big)_1^k$, for $k = 0, 1, 2, 3, 4$. Show that for $k = 1, 3$ we get 0, for $k = 0$ we get 1, for $k = 2$ we get $\frac{1}{d}$, and for $k = 4$ we get at most $\frac{3}{d^2}$.

(c) Show that for all positive odd k we have

$$\mathbb{E}\Big(\Big(\sum_{j=1}^{n} x_j W_{1j}\Big)^k\Big) = 0\,,$$

and for all positive even k we have

$$0 \le \mathbb{E}\Big(\Big(\sum_{j=1}^{n} x_j W_{1j}\Big)^k\Big) \le \frac{k-1}{d}\mathbb{E}\Big(\Big(\sum_{j=1}^{n} x_j W_{1j}\Big)^{k-2}\Big)\,.$$

Hint: consider expanding out the kth power. When we choose a term from the first bracket, to get something nonzero we must choose the same term in one of the other $k-1$ brackets, and this is the factor $k-1$ in the inequality.

(d) Calculate $\mathbb{E}\big(Y^k\big)$, where Y is distributed as $N(0, \frac{1}{d})$, for each integer $k \ge 0$. You should leave your answer in terms of $\mathbb{E}\big(Y^{k-2}\big)$ for $k \ge 2$.

(e) Prove that for all integer $k \ge 0$ we have

$$0 \le \mathbb{E}\Big(\Big(\sum_{j=1}^{n} x_j W_{1j}\Big)^k\Big) \le \mathbb{E}\Big(Y^k\Big)\,,$$

where Y is distributed as $N(0, \frac{1}{d})$.

(f) Prove that $\mathbb{E}\Big(\exp\Big(t\big(\sum_{j=1}^{n} x_j W_{1j}\big)^2\Big)\Big) \le \big(1 - \frac{2t}{d}\big)^{-1/2}$ holds whenever $0 \le \frac{2t}{d} < 1$.

(g) Prove the Johnson-Lindenstrauss Projection Theorem for the binary random matrix W: i.e. that for any fixed $\mathbf{x} \in \mathbb{R}^n$ (not necessarily with $\|\mathbf{x}\|_2 = 1$), with probability at least $1 - 2e^{-\varepsilon^2 d/8}$ we have

$$(1-\varepsilon)\|\mathbf{x}\|_2 \le \|W\mathbf{x}\|_2 \le (1+\varepsilon)\|\mathbf{x}\|_2 \,.$$

Hint: the upper bound should work much as for the Gaussian random matrix, but for the lower bound you need to show that if $0 < t < \frac{d}{4}$ then

$$\mathbb{E}\Big(\exp\Big(-t\big(\sum_{j=1}^n x_j W_{1j}\big)^2\Big)\Big) \le 1 - \tfrac{t}{1!d} + \tfrac{3t^2}{2!d^2} \,.$$

To do this, use the fact that the Taylor series is alternating.

The Johnson-Lindenstrauss Theorem for binary matrices now follows by the same proof as in the lecture: so we don't need to repeat that.

Solution to Exercise 6:

(a) This is immediate from the definition: $\mathbb{E}(W_{ij}) = \frac{1}{2}\cdot\frac{-1}{\sqrt{d}} + \frac{1}{2}\cdot\frac{1}{\sqrt{d}} = 0$ and $\mathbb{E}(W_{ij}^2) = \frac{1}{2}\cdot\frac{1}{d} + \frac{1}{2}\cdot\frac{1}{d} = \frac{1}{d}$.

(b) This part of the exercise is 'shut up and calculate' (due to David Mermin), i.e. figure out a few things by explicit calculation before trying to find a pattern.

For $k = 0$ we (obviously!) get 1, and for $k = 1$ we get 0 by linearity of expectation. For $k = 2$, we have

$$\mathbb{E}\Big(\Big(\sum_{j=1}^n x_j W_{1j}\Big)^2\Big) = \mathbb{E}\Big(\sum_{j=1}^n x_j^2 W_{1j}^2 + 2\sum_{j<j'} x_j x_{j'} W_{1j} W_{1j'}\Big) = \tfrac{1}{d}\sum_{j=1}^n x_j^2 = \tfrac{1}{d} \,.$$

Here we see a useful observation. The expectation of the 'cross terms' disappears, because we can split $\mathbb{E}(W_{1j}W_{1j'}) = \mathbb{E}(W_{1j})\cdot\mathbb{E}(W_{1j'})$ as the two random variables are independent, and the first term is zero (so is the second). This is going to happen every time that we have a term with W_{1j}^k in it where k is an odd integer. Going further, our random variable W_{1j} doesn't just have expectation zero: it is symmetric about zero with sign reversed, so all the higher moments are zero too. (Explicitly, if k is odd then $\mathbb{E}(W_{1j})^k = \frac{1}{2}\cdot(-1/\sqrt{d})^k + \frac{1}{2}\cdot(1/\sqrt{d})^k = 0$.) So if we have a term which *doesn't* vanish in $\mathbb{E}\big(\big(\sum_{j=1}^n x_j W_{1j}\big)^k\big)$ then this means we pick each term $x_j W_{1j}$ an even number of times, and when we take the expectation we will get a product of x_j terms multiplied by $d^{-k/2}$.

For $k = 3$, when we expand the brackets $\big(\sum_{j=1}^n x_j W_{1j}\big)^3$ we can't pick every term an even number of times, since we have an odd number of brackets in total. So the expectation will be zero by the observation above, and this goes for every odd k.

For $k = 4$, we get

$$\mathbb{E}\Big(\Big(\sum_{j=1}^n x_j W_{1j}\Big)^4\Big) = d^{-2}\sum_{j=1}^n x_j^4 + d^{-2}\binom{4}{2}\sum_{j<j'} x_j^2 x_{j'}^2$$

again using the above observation. We could go further and write out more such things with k even. What we should notice, though, is all these expectations are either zero or strictly positive: no matter what the coordinates of $\mathbf{x}$ are, we have no negative expectations.

As we've written it, the fourth moment is given by a not particularly intelligible formula—meaning, we can't really see what is going on. However, something we can notice is:

$$\mathbb{E}\Big(\Big(\sum_{j=1}^{n} x_j W_{1j}\Big)^4\Big) \leq 3d^{-2}\Big(\sum_{j=1}^{n} x_j^2\Big)^2 = 3d^{-2}\,.$$

At the moment this is just a numerical observation (expand out the square and check it's true!) but the point of the next part is to explain it. The point of this is that a 'nice' upper ound that doesn't depend on the coordinates of $\mathbf{x}$ is going to be easier to work with than a painful exact formula.

(c) Both the odd and even k parts of this rely on the observation from above that when we expand out the brackets in

$$\mathbb{E}\Big(\Big(\sum_{j=1}^{n} x_j W_{1j}\Big)^2\Big)$$

any term which has an odd power of any W_{1j} has expectation zero. When k is odd, this is obviously going to be every single term: this justifies that for odd k the expectation is zero. Furthermore, any term where all the powers are even is a non-negative random variable.

Suppose now k is even. The lower bound, that the expectation is non-negative, is obvious: we're taking the expectation of a non-negative random variable. Let's look at the upper bound.

Following the Hint, consider expanding out the brackets, and suppose that we pick a term $x_j W_{1j}$ from the first bracket. Expanding out the rest of the brackets, if we don't pick $x_j W_{1j}$ in any other bracket, then the term we're writing out in the expansion has expectation zero (because the power of W_{1j} is 1, which is odd) and so doesn't contribute to the sum. So we only care about the expansions where there is at least one other bracket number s, with $2 \leq s \leq k$, where we pick the $x_j W_{1j}$ term.

Now all the terms we are looking at are in the product

$$x_j^2 W_{1j}^2 \Big(\sum_{i=1}^{n} x_i W_{1i}\Big)^{k-2} = x_j^2 d^{-1}\Big(\sum_{i=1}^{n} x_i W_{1i}\Big)^{k-2}\,.$$

Here we used a convenient property of the binary random variable: $W_{1j}^2 = d^{-1}$ holds surely. This is *not* true for any other distribution, and dealing with any other distribution needs more work! What we did here was to use it to substitute in a value for W_{1j}^2 without it affecting the rest of the random variable, even though W_{1j} appears again in the brackets.

This means that the terms we've looked at—all terms where we pick $x_j W_{1j}$ in the first bracket and again in the sth—contribute

$$x_j^2 d^{-1}\mathbb{E}\Big(\Big(\sum_{i=1}^{n} x_i W_{1i}\Big)^{k-2}\Big)$$

to the expectation we're trying to calculate. So the contribution of all terms where we pick $x_j W_{1j}$ in the first bracket is at most

$$(k-1)\cdot x_j^2 d^{-1}\mathbb{E}\Big(\Big(\sum_{i=1}^{n} x_i W_{1i}\Big)^{k-2}\Big)$$

because there are $k-1$ choices for s. This is an upper bound, because we counted some terms multiple times (where we chose $x_j W_{1j}$ in four or more brackets!); here we're using that all these terms are non-negative.

Finally, summing over the choice of j, we have

$$\mathbb{E}\Big(\Big(\sum_{j=1}^n x_j W_{1j}\Big)^k\Big) \le \sum_{j=1}^n (k-1) x_j^2 d^{-1} \mathbb{E}\Big(\Big(\sum_{i=1}^n x_i W_{1i}\Big)^{k-2}\Big) = \tfrac{k-1}{d}\mathbb{E}\Big(\Big(\sum_{i=1}^n x_i W_{1i}\Big)^{k-2}\Big),$$

where we used the fact $\sum_{j=1}^n x_j^2 = 1$.

Here is an alternative way to see why we get the claimed inequality. Consider any term in $\big(\sum_{j=1}^n y_j\big)^k$. (I'm writing y_i not $x_i W_{1i}$ just to save space.) Any term is of the form $\prod_{j=1}^n y_j^{\alpha_j}$, where $\sum_j \alpha_j = k$, multiplied by some number C_1: the number is the multinomial coefficient $\binom{k}{\alpha_1,\dots,\alpha_n}$. This number is symmetric in the α_i (i.e. the order doesn't matter) and we can remove all the zero terms: without loss of generality, we can consider $\binom{k}{\alpha_1,\dots,\alpha_\ell}$ where each α_i is a positive integer.

We are only interested in terms where all the α_i are even: but for this part, what we'll actually use is that none of them is equal to 1.

We have the usual recurrence relation

$$C_1 = \binom{k}{\alpha_1,\dots,\alpha_\ell} = \binom{k-1}{\alpha_1-1,\alpha_2,\dots,\alpha_\ell} + \binom{k-1}{\alpha_1,\alpha_2-1,\dots,\alpha_\ell} + \cdots + \binom{k-1}{\alpha_1,\alpha_2,\dots,\alpha_\ell-1},$$

which comes from the following: we can get our desired term by choosing any one of $y_1,\dots,y_\ell$ from the first bracket $(y_1+\cdots+y_n)$; then we need to choose the right numbers of the remaining y_i from the remaining $k-1$ brackets.

What we are trying to compare this to is $(y_1^2+\cdots+y_n^2)\big(\sum_{j=1}^n y_j\big)^{k-2}$. Here the coefficient C_2 of $\prod_{j=1}^n y_j^{\alpha_j}$ is given by a rather similar sum:

$$C_2 = \binom{k-2}{\alpha_1-2,\alpha_2,\dots,\alpha_\ell} + \binom{k-2}{\alpha_1,\alpha_2-2,\dots,\alpha_\ell} + \cdots + \binom{k-2}{\alpha_1,\alpha_2,\dots,\alpha_\ell-2},$$

This comes from similar logic: we need to choose one of $y_1^2,\dots,y_\ell^2$ from the first bracket, then the right numbers of the remaining y_i from the $k-2$ brackets left.

We want to show $C_1 \le (k-1)C_2$. It's enough to show we have a corresponding inequality term by term:

$$\binom{k-1}{\alpha_1-1,\alpha_2,\dots,\alpha_\ell} \le (k-1)\binom{k-2}{\alpha_1-2,\alpha_2,\dots,\alpha_\ell}$$

and so on for the other $\ell-1$ terms. But again, because the multinomial is symmetric in the bottom coefficients, it's enough to show this one statement holds. We can just write out the multinomials:

$$\binom{k-1}{\alpha_1-1,\alpha_2,\dots,\alpha_\ell} = \frac{(k-1)!}{(\alpha_1-1)!\alpha_2!\dots\alpha_\ell!} = \frac{k-1}{\alpha_1-1}\cdot\frac{(k-2)!}{(\alpha_1-2)!\alpha_2!\dots\alpha_\ell!} = \frac{k-1}{\alpha_1-1}\binom{k-2}{\alpha_1-2,\alpha_2,\dots,\alpha_\ell}.$$

Now since $\alpha_1 \ge 2$, we have the inequality we wanted: $\binom{k-1}{\alpha_1-1,\alpha_2,\dots,\alpha_\ell} \le (k-1)\binom{k-2}{\alpha_1-2,\alpha_2,\dots,\alpha_\ell}$, (actually with equality exactly when $\alpha_1 = 2$) and summing up $C_1 \le (k-1)C_2$.

This proves

$$\mathbb{E}\Big(\Big(\sum_{i=1}^n x_i W_{1i}\Big)^k\Big) \le (k-1)\mathbb{E}\Big(\Big(\sum_{i=1}^n x_i^2 W_{1i}^2\Big)\Big(\sum_{i=1}^n x_i W_{1i}\Big)^{k-2}\Big)$$

because all the terms in the expectation are non-negative, only the terms where all the powers are even contribute something nonzero, and for those terms we just showed the coefficient on the left is at most the coefficient on the right.

Now, for every point in our probability space we have $\sum_{i=1}^n x_i^2 W_{1i}^2 = \frac{1}{d}$. (This is where we use that we're dealing with this special random variable.) So we can simply replace this sum by $\frac{1}{d}$:

$$\mathbb{E}\Big(\Big(\sum_{i=1}^n x_i W_{1i}\Big)^k\Big) \le (k-1)\mathbb{E}\Big(\tfrac{1}{d}\Big(\sum_{i=1}^n x_i W_{1i}\Big)^{k-2}\Big),$$

which is what we wanted to prove.

If you prefer, you can do this last bit in terms of independence. Let $P = \sum_{i=1}^n x_i^2 W_{1i}^2$ and $Q = \Big(\sum_{i=1}^n x_i W_{1i}\Big)^{k-2}$. Now P and Q are independent random variables (even if they don't look like it!): $\sigma(P) = \{\emptyset, \Omega\}$ which is by definition independent of all σ-algebras on Ω. (Yes, P is even independent of P.) So we can use multiplicativity: $\mathbb{E}(PQ) = \mathbb{E}(P)\mathbb{E}(Q)$ which is what we wanted.

(d) This is an exercise in integration; we use the Law of the Unconscious Statistician (LOTUS) to write

$$\mathbb{E}\big(Y^k\big) = \int_{x=-\infty}^{\infty} x^k \tfrac{\sqrt{d}}{\sqrt{2\pi}} e^{-x^2 d/2}\,\mathrm{d}x$$

and then we need to calculate this. For $k = 0$ we get 1: this is the Gaussian integral. For $k = 1$ we get 0 because the integrand is symmetric about 0 and reverses sign.

For $k \ge 2$ we use integration by parts: one part is x^{k-1}, whose derivative is $(k-1)x^{k-2}$ and the other is $xe^{-x^2 d/2}$, whose integral is $d^{-1}e^{-x^2 d/2}$. The limit as $x \to \infty$ or as $x \to \infty$ of $x^{k-2}e^{x^2 d/2}$ is zero, so we get

$$\mathbb{E}\big(Y^k\big) = \int_{x=-\infty}^{\infty} (k-1)\cdot \tfrac{\sqrt{d}}{\sqrt{2\pi}} x^{k-2} d^{-1} e^{-x^2 d/2}\,\mathrm{d}x = \tfrac{k-1}{d}\mathbb{E}\big(Y^{k-2}\big)$$

where the last equality again uses the LOTUS.

(e) From (c) and (d), we have for all odd $k \ge 1$ the equality

$$0 = \mathbb{E}\Big(\Big(\sum_{j=1}^n x_j W_{1j}\Big)^k\Big) = \mathbb{E}\big(Y^k\big),$$

as required. For $k = 0, 2$, we also have equality as required, because we calculated exactly both terms; we got 1 and d^{-1} respectively for the binary random variables in (b), and from (d) we get the same for the Gaussian.

We now perform induction on the even integers, with $k = 2$ as the base case. For even $k \ge 4$, we have

$$0 \le \mathbb{E}\Big(\Big(\sum_{j=1}^n x_j W_{1j}\Big)^k\Big) \le \tfrac{k-1}{d}\mathbb{E}\Big(\Big(\sum_{j=1}^n x_j W_{1j}\Big)^{k-2}\Big) \le \tfrac{k-1}{d}\mathbb{E}\big(Y^{k-2}\big) = \mathbb{E}\big(Y^k\big),$$

where the first inequality is obvious (we're taking the expectation of a non-negative random variable), the second is (c), the third is the induction hypothesis—our desired inequality holds for $k-2$—and the final equality is (d). This proves the induction step, so we are done by the Principle of Induction.

(f) The idea here is simply to expand out the exponential as a Taylor series. Let $X = \sum_{i=1}^n x_j W_{1j}$. Observe that this random variable is at most $\sum_{i=1}^n x_j d^{-1/2} \leq \sqrt{n/d}$, where the inequality is Jensen's inequality. We want to prove $\mathbb{E}(e^{tX^2}) \leq \left(1 - \frac{2t}{d}\right)^{-1/2}$ for $\frac{2t}{d} < 1$.

Since the Taylor series for the exponential converges pointwise on $\mathbb{R}$, we have

$$e^{tX^2} = 1 + \sum_{k=1}^{\infty} \tfrac{t^k X^{2k}}{k!}, \quad \text{and} \quad f_m = 1 + \sum_{k=1}^{m} \tfrac{t^k X^{2k}}{k!}$$

is therefore a sequence of functions converging pointwise to e^{tX^2}. Because $t \geq 0$, the functions are monotone increasing pointwise, so by the Monotone Convergence Theorem we have

$$\mathbb{E}\big(e^{tX^2}\big) = \lim_{m\to\infty} \mathbb{E}(f_m) = \lim_{m\to\infty} \mathbb{E}\big(1 + \sum_{k=1}^{m} \tfrac{t^k X^{2k}}{k!}\big) = \lim_{m\to\infty} 1 + \sum_{k=1}^{m} \tfrac{t^k}{k!}\mathbb{E}\big(X^{2k}\big),$$

where the MCT is the first equality, the second is the definition of f_m, and the third is linearity of expectation (we were careful to get to a finite sum here!). But now we can substitute in our upper bounds from (e):

$$\mathbb{E}\big(e^{tX}\big) \leq \lim_{m\to\infty} 1 + \sum_{k=1}^{m} \tfrac{t^k}{k!}\mathbb{E}\big(Y^{2k}\big) = \mathbb{E}\big(e^{tY^2}\big) = \left(1 - \tfrac{2t}{d}\right)^{-1/2},$$

where we should be clear that the left-side expectation is over the probability space of our binary random matrices, while the middle expectations are over a completely different probability space supporting a Gaussian random variable $Y \sim N(0, d^{-1})$. If you prefer, the right-side expectation is simply a number we can calculate from (d). The first equality is the Monotone Convergence Theorem, and the second is Exercise 4.

We should notice that we needed $t \geq 0$ here to justify the inequality in the final display equation: if $t < 0$ then some of the inequalities from (e) go in the wrong direction.

(g) We have $\mathbb{E}\big(e^{t\|W\mathbf{x}\|_2^2}\big) \leq \left(1 - \frac{2t}{d}\right)^{-d/2}$ because each of the d entries of $W\mathbf{x}$ is independently distributed as in (f): that is

$$\mathbb{E}\big(e^{t\|W\mathbf{x}\|_2^2}\big) = \mathbb{E}\big(e^{tX_1^2}\big) \cdots \mathbb{E}\big(e^{tX_d^2}\big)$$

where X_i is the ith entry of $W\mathbf{x}$. We can use Markov's inequality much as in the proof of Hoeffding's inequality: for any $0 < t < \frac{d}{2}$, we have

$$\mathbb{P}\big(\|W\mathbf{x}\|_2^2 \geq (1+\varepsilon)^2\big) = \mathbb{P}\Big(e^{t\|W\mathbf{x}\|_2^2} \geq e^{t(1+\varepsilon)^2}\Big) \leq \frac{\mathbb{E}\Big(e^{t\|W\mathbf{x}\|_2^2}\Big)}{e^{t(1+\varepsilon)^2}} \leq \frac{\left(1 - \frac{2t}{d}\right)^{-d/2}}{e^{t(1+\varepsilon)^2}}$$

where the first inequality is Markov's inequality and the second is substituting our estimate from (f). Now we can copy from the previous proof: we substitute $t = \frac{d}{2}\frac{2\varepsilon+\varepsilon^2}{(1+\varepsilon)^2}$, and (because it's verbatim the same formula) we get

$$\frac{\left(1 - \frac{2t}{d}\right)^{-d/2}}{e^{t(1+\varepsilon)^2}} \leq e^{-\varepsilon^2 d/8}.$$

This is the upper bound we wanted.

Now we want to do the lower bound. We can try to follow the same strategy as before: for any $t > 0$ we have

$$\mathbb{P}\big(\|W\mathbf{x}\|_2^2 \le (1-\varepsilon)^2\big) = \mathbb{P}\Big(e^{-t\|W\mathbf{x}\|_2^2} \ge e^{-t(1-\varepsilon)^2}\Big) \le \frac{\mathbb{E}\Big(e^{-t\|W\mathbf{x}\|_2^2}\Big)}{e^{-t(1-\varepsilon)^2}},$$

where the inequality in the probability 'flipped' because $z \to e^{-tz}$ is a monotone decreasing function for $t > 0$.

However, we can't any more just plug in a formula for the moment generating function: the inequality from (f) isn't valid for negative inputs. We'll need a counterpart to the bound in (f) for negative t.

Following the Hint, we'll look at the first three terms in the Taylor series expansion of e^{-x}. We hope that for $x \ge 0$ we can write $e^{-x} \le 1 - x + \frac{1}{2}x^2$.

Indeed, if $0 \le x \le 1$, then the full Taylor series is an alternating series whose terms are monotone decreasing in absolute value (and tend to 0), so by a standard result every even term is an upper bound for the limit; in particular $e^{-x} \le 1 - x + \frac{1}{2}x^2$. And if $x \ge 1$ then the derivative of $1 - x + \frac{1}{2}x^2$ is $-1 + x \ge 0$, while the derivative of e^{-x} is $-e^{-x} < 0$, so by the Mean Value Theorem we conclude $e^{-x} \le 1 - x + \frac{1}{2}x^2$ holds for all $x \ge 0$.

By monotonicity of the integral, and since $tX^2 \ge 0$, we have

$$\mathbb{E}\big(e^{-tX^2}\big) \le \mathbb{E}\Big(1 - tX^2 + \tfrac{1}{2}t^2X^4\Big) = 1 - t\mathbb{E}(X^2) + \tfrac{1}{2}t^2\mathbb{E}(X^4) \le 1 - \tfrac{t}{d} + \tfrac{3t^2}{2d^2}$$

where we used our exact calculation of $\mathbb{E}(X^2)$ and upper bound on $\mathbb{E}(X^4)$ from (b).

(Another way to get to this is to use the DCT, with a suitable dominating function, to write $\mathbb{E}\big(e^{-tX^2}\big) = 1 + \sum_{i=1}^{\infty} \frac{(-t)^i}{i!}\mathbb{E}(X^{2i})$, and then argue that the terms in this infinite series have decreasing absolute value so that $1 - t\mathbb{E}(X^2) + \frac{t^2}{2}\mathbb{E}(X^4)$ is an upper bound. This requires slightly more calculation but works too.)

As before, we get for $t > 0$:

$$\mathbb{E}\big(e^{-t\|W\mathbf{x}\|_2^2}\big) \le \big(1 - \tfrac{t}{d} + \tfrac{3t^2}{2d^2}\big)^d$$

and we can substitute this in to get

$$\mathbb{P}\big(\|W\mathbf{x}\|_2^2 \le (1-\varepsilon)^2\big) \le \frac{\big(1 - \frac{t}{d} + \frac{3t^2}{2d^2}\big)^d}{e^{-t(1-\varepsilon)^2}} = \big(1 - \tfrac{t}{d} + \tfrac{3t^2}{2d^2}\big)^d e^{t(1-\varepsilon)^2}. \tag{1}$$

It's not easy to see how to minimise this precisely, but we also do not need to know an exact minimum. Let's use another fact about e^{-x}, namely that for all $0 \le x \le 1$ we have $1 - x \le e^{-x}$. As before, this follows since the Taylor series is alternating. Using this, we get (provided we choose t sufficiently small, which we will!)

$$\big(1 - \tfrac{t}{d} + \tfrac{3t^2}{2d^2}\big)^d e^{t(1-\varepsilon)^2} \le \exp\Big(\big(-\tfrac{t}{d} + \tfrac{3t^2}{2d^2}\big)d + t(1-\varepsilon)^2\Big)$$

and minimising this means minimising the quadratic in t: $\frac{3}{2d}t^2 - t + t(1-\varepsilon)^2 = \frac{3}{2d}t^2 - 2\varepsilon t + \varepsilon^2 t$. Again, we don't really need to get exactly the minimum of this. We just need to find a choice

for t which gets a value below $-\varepsilon^2 d/8$. Let's just plug in $t = \frac{1}{8}\varepsilon d$ and hope that the negative term $2\varepsilon t$ will be sufficiently small (i.e. we hope the factor of 2 will kill the positive terms). Indeed, this works: we get $d\big(\frac{3\varepsilon^2}{128} - \frac{\varepsilon^2}{4} + \frac{\varepsilon^3}{8}\big) \leq d\varepsilon^2\big(\frac{3}{128} - \frac{1}{4} + \frac{1}{16}\big) < -\frac{1}{8}\varepsilon^2 d$ using our assumption $0 < \varepsilon \leq \frac{1}{2}$.

Since this choice of t is (as required!) small enough, we now proved

$$\mathbb{P}\big(\|W\mathbf{x}\|_2^2 \leq (1-\varepsilon)^2\big) \leq e^{-\varepsilon^2 d/8}$$

as we wanted. By the union bound, the probability that either the upper or lower bound on $\|W\mathbf{x}\|_2^2$ fails is at most $2e^{-\varepsilon^2 d/8}$, and this proves the Johnson-Lindenstrauss Projection Theorem for binary matrices: with probability at least $1 - 2e^{-\varepsilon^2 d/8}$, for any fixed $\mathbf{x}$ and the random binary matrix W we have

$$(1-\varepsilon)\|\mathbf{x}\|_2 \leq \|W\mathbf{x}\|_2 \leq (1+\varepsilon)\|\mathbf{x}\|_2 \,.$$

Of course, what you *should* do if you are faced with this kind of ugly minimisation problem is not to throw algebra at it (as above) but to get a computer to help guess. As soon as you get to (1), you can guess that the solution you're after is of the form $t = dx$ for some x (we want to get a linear dependence on d for both terms!). So plug that in to a computer and let it graph what you get over x, for some example values of ε. You'll quickly see that the 'best' choice of x seems to go linearly in ε to 0: OK, then let's update the guess to $t = \varepsilon d y$ for some y. Eyeballing the result, there will be lots of choices of y that look good; $\frac{1}{8}$ is one, and you can then see if it really works.

This 'binary' Johnson-Lindenstrauss Theorem is something friendly to data scientists. The factor $\frac{1}{\sqrt{d}}$ is not something one wants to work with, but it simply multiplies all the distances. The data scientist will simply use a matrix whose entries are uniformly and independently chosen in $\{-1, 1\}$ and remember that distances have been approximately preserved, up to a scaling factor $\sqrt{d}$: and scaling doesn't really affect the 'structure' of the data. So they will still be working with integer vectors in our Amazon example.

In fact, a bit more is true (though this isn't used much, so far as I know). In practice, no customer ever buys any significant fraction of the items available on Amazon: every customer corresponds to a *s-sparse* vector, where there are only at most s, non-zero entries.

In this setting, if you're willing to pay a price (the dimension d you project to needs to be bigger), you can actually recover each customer's original purchases from the projected vector. Cheating somewhat, let's suppose customers buy at most one of each item.

To do this, we let the S of the Johnson-Lindenstrauss Theorem be the set of vectors $\mathbf{v}$ all of whose entries are either 0 or 1, and with at most s ones; and we can let $\varepsilon = 0.5$. This set has size

$$\sum_{i=0}^{s} \binom{n}{i} \leq s\binom{n}{s} \leq s\big(\tfrac{en}{s}\big)^s \,.$$

(The final bound is a standard upper bound on binomial coefficients: to prove it use Stirling's approximation for the factorial.) We need d large enough to allow for this: for example $d = 8\varepsilon^{-2} s \log n$ works. Now given $W\mathbf{v}$, we simply look at all sparse vectors with 0 and 1 entries until we find $\mathbf{v}'$ such that $W\mathbf{v}' = W\mathbf{v}$. We claim that in fact we must have $\mathbf{v} = \mathbf{v}'$: if

not, then $\|\mathbf{v}-\mathbf{v}'\| \geq 1$, so Johnson-Lindenstrauss guarantees $\|W(\mathbf{v}-\mathbf{v}')\|_2 \geq (1-\varepsilon)\cdot 1 = \frac{1}{2}$. But by linearity we have $W(\mathbf{v}-\mathbf{v}') = 0$.

With a bit of thought, this can actually be extended to deal with any collection of s-sparse vectors, not necessarily even with integer entries (though it's easier!). And there is actually a much faster reconstruction procedure than the 'brute force check' outlined above (which in practice you can't even think of running on a computer).

The Johnson-Lindenstrauss Theorem (for much the same reason as above) also provides the 'probabilistic' part of proving that something called 'compressive sensing' works.

If we have a 'simple' image we want to reconstruct—such as an MRI scan of an injured person—we can reasonably expect that this image contains 'not much' actual data. For example, we can try to find where the colour in the image changes, and expect that there are not many such places: if we know where and how the colour changes, that's enough to reconstruct the whole image.

This is hard to visualise in 2 dimensions (because I'm cheating, partly! in reality this uses the 2-dimensional discrete Fourier transform) but in 1 dimension we can imagine a simple image would be n pixels in a row—a vector—but where there are only at most s colour changes as we go along the row, for some s much smaller than n.

We could just replace the image with a much simpler vector $\mathbf{v}$, where the value at each pixel is the amount the colour changed between the previous pixel and the current one. This is 'simpler' because most of the entries are 0: our vector will be 's-sparse', meaning there are at most s non-zero entries, because there were at most s colour changes.

Our MRI machine can be set to scan the colour change at each pixel one after another. But it can also be set to find the value of any linear combination of colour changes in a single scan. (That might seem unreasonable, but actually it isn't; any given linear combination turns out to take about as much MRI time as one single colour change.)

So we can take a Gaussian random matrix A as in the Johnson-Lindenstrauss Theorem and directly find the d entries of $A\mathbf{v}$: that's just d linear combinations to scan one after another.

Much as above: we were exceptionally lucky and wanted to reconstruct a black-and-white image, the entries of $\mathbf{v}$ would be either 0 or 1; if we choose d sensibly in terms of s, we can let S be the entire set of s-sparse vectors in the Johnson-Lindenstrauss Theorem, and it follows that just knowing $A\mathbf{v}$ is enough data to reconstruct $\mathbf{v}$ exactly: there can't be two s-sparse vectors $\mathbf{v}$ and $\mathbf{v}'$ such that $A\mathbf{v} = A\mathbf{v}'$ because that would contradict the theorem's conclusion that distances between all pairs are approximately preserved.

In fact, there is a 'trick' to make this work even if we don't know that the entries of $\mathbf{v}$ are always 0 or 1, but only that there are few non-zero entries. And in fact, although the previous paragraph suggests that 'reconstruction' means checking all s-sparse vectors—of which there are a lot, so that this wouldn't be practical!—to see which one works, there is another 'trick' that uses linear programming to do the job much faster.

This is, more or less, compressive sensing: in some situations one can do this 'scan a random linear combination' thing, and if additionally you want to reconstruct a 'sparse' image, you can, and you can do so without excessive computation. In terms of MRI scanning,

it reduces hugely the amount of time needed to generate an image; Siemens, for instance, claim a factor 20 speed-up in their commercial products; the particular example they give is time for a cardiac MRI scan (during which the patient must hold their breath and stay still) going from over 6 minutes to around 20 seconds—this means going from needing the patient to repeatedly hold breath and try to stay otherwise still for 6 minutes to (for most people) being able to do it with just a single breath-hold.